

FirstModuAR: Open-Source Modular Augmented Reality Interfaces for First Responders

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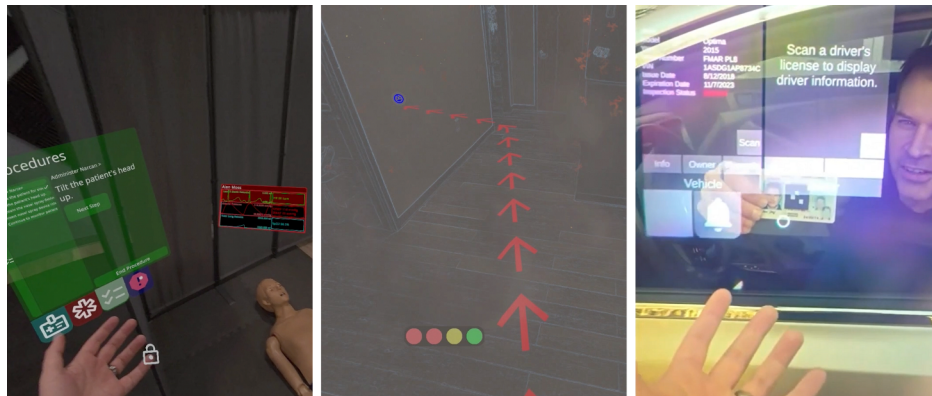


Figure 1: FirstModuAR Emergency Response Scenarios. Left: an EMS scenario showing an augmented on-demand display with procedure guidance and an augmented vitals monitor above a patient simulator. Center: a fire rescue scenario with simulated smoke and fire, showing augmented edges, HUD air level, and an augmented path to the victim's location. Right: A law enforcement traffic stop scenario, showing the same information display with vehicle data. The law enforcement scenario is depicted on the Magic Leap 2 VR HMD, while the other scenarios are depicted on the Meta Quest 2.

ABSTRACT

First responders operate in high-pressure environments where decisions must be made quickly and accurately. However, the increasing availability of real-time data presents a double-edged sword; while potentially valuable, the sheer volume of information can lead to cognitive overload, compromising situational awareness and performance. The FirstModuAR project seeks to address these challenges by developing modular Augmented Reality (AR) interfaces specifically tailored to public safety. By working closely with first responders, the project focuses on creating reusable AR modules that improve situational awareness and decision-making across diverse scenarios. This paper provides an overview of the system architecture, the modular design approach, and the integration of AR solutions into realistic public safety scenarios.

Index Terms: Augmented reality, modular user interfaces, first responders, public safety.

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1 INTRODUCTION

In public safety contexts, such as firefighting, emergency medical services (EMS), and law enforcement, responders are increasingly equipped with tools that provide real-time data [1]. While these tools have immense potential, improper presentation of data can overwhelm users, reducing their ability to assess and respond effectively to their surroundings. Augmented reality (AR) offers a unique opportunity to present data intuitively, overlaying critical information within the responder's field of view to augment situational awareness without impeding physical actions [2].

The *FirstModuAR*¹ project is an effort toward the design and development of tested and proven public safety AR user interfaces (UIs) through a modular and adaptable approach. Unlike traditional AR systems, which often focus on isolated use cases, the modular approach enables the creation of reusable interface components that can be adapted across multiple scenarios. This design ensures scalability, reduces development overhead, and addresses the evolving needs of public safety operations. FirstModuAR contributes toward accelerating AR adoption in public safety by integrating demonstrably usable and effective AR UIs.

2 SYSTEM ARCHITECTURE

The system architecture of *FirstModuAR* balances modularity with real-world usability, integrating seamlessly with current AR hardware while maintaining flexibility for future advancements. Central to this system is a hardware-agnostic development framework, pri-

¹FirstModuAR Unity modules with source code are licensed under Apache 2.0 and available at <https://fmar.nextgeninteractions.com>.

marily implemented in Unity, which ensures compatibility across various AR platforms. FirstModuAR is currently configured to build applications for the Magic Leap 2 and Meta Quest 3 head-mounted displays (HMDs). Other current or future HMDs can be integrated into FirstModuAR by instantiating device-specific providers with little to no need to change the implementation of existing modules. This capability ensures that the modules can be transferred to other devices as AR hardware evolves.

An important factor considered in the design of the FirstModuAR architecture was the interaction modality. During emergency calls, the undivided attention of first responders should be directed at the public safety operation. Because of that, FirstModuAR utilizes a free-hand gesture-based interaction interface that allows intuitive controls without the need for external controllers. Networking capabilities have also been integrated to enable real-time data sharing among multiple responders, increasing situational awareness and allowing communication across first responders.

The modularity of the software framework ensures that each AR module operates as an independent unit, capable of being customized and reconfigured for specific operational needs. For instance, a module designed for navigation can be repurposed across firefighting, EMS, and law enforcement scenarios without significant redevelopment efforts. FirstModuAR also supports message passing across different modules, enabling the synchronization of data and triggering events from one module to another. This is useful, for example, for alerts to be triggered by data coming from other modules, such as an augmented patient monitor generating an alert due to a critical threshold being surpassed.

3 DESIGNED MODULES

FirstModuAR currently offers a library of modules designed in collaboration with first responders and can be integrated into multiple scenarios.

The *Information Display* module provides a canvas for in-context data visualization by first responders. It has three presentation modes: on-demand, environment-anchored, and body-anchored. The on-demand display can be summoned by an open-palm-raising gesture and contains tabs that present different types of information. Figure 1 shows the on-demand display in a Meta Quest 3 build of an EMS scenario (left) and in a Magic Leap 2 build of a traffic stop scenario (right). A configuration of the information display module anchored in the environment is shown in Figure 1 left in the form of a patient vitals monitor. Finally, the module can be attached to a body part, such as the user's wrist, as shown in the air tank gauge visualization (Figure 2).



Figure 2: Body-anchored information display module showing an air gauge visualization

Similarly, the *Map and Navigation Module* supports spatial awareness by offering real-time location tracking and route guidance, including dynamic waypoints and annotated maps for enhanced coordination.

The *Notification System* module, which provides non-intrusive alerts through peripheral visual effects and auditory cues, can be triggered by other modules, such as the air gauge, as soon as the value goes below a preset threshold.

Two modules provide navigation cues. The *Environment Overlay* module (Figure 1 left) shows arrows on the ground leading to a target location. The *Minimap* module gives a visualization of the building plan and adds information such as current path overlays and the location of other firefighters and points of interest.

Other modules include *Object Highlighting*, which allows responders to identify and track critical objects in their environment, and *Edge Detection*, which can improve environment awareness in low visibility environments.

Finally, the *Quick Access Radial Menu* module can be used to trigger *Points of Interest Markers*, such as hazard indicators, and the *Procedure Visualization* module (Figure 1 left) can be used to guide first responders).

4 INTEGRATED SCENARIOS

The modular AR solutions developed in FirstModuAR are demonstrated in scenarios replicating critical public safety operations. These scenarios serve as testing grounds for evaluating the effectiveness of the AR modules under realistic conditions.

The search and rescue scenario involves a firefighting operation in a smoke-filled building where AR is used to augment visibility and navigation. The system overlays augmented navigational cues and hazard indicators, enabling firefighters to locate and rescue trapped individuals despite reduced visibility.

In the EMS scenario, responders monitor two patients during a suspected opioid overdose. The AR system provides real-time updates on patient vitals and alerts responders when intervention is needed. For instance, the system guides responders in administering Narcan when a patient's vitals drop below safe thresholds, ensuring rapid and effective intervention.

The law enforcement traffic stop scenario includes enhanced situational awareness through license plate recognition and database integration. This allows officers to receive critical information about vehicles and drivers, enhancing decision-making and reducing uncertainty during stops.

5 CONCLUSION

The FirstModuAR project demonstrates the transformative potential of modular AR systems in public safety. By addressing the specific challenges first responders face, the project delivers reusable, adaptable solutions that enhance situational awareness, streamline decision-making, and improve overall safety. The integrated scenarios underscore the practicality and effectiveness of these systems, paving the way for broader adoption of AR technologies in public safety. Future efforts will focus on expanding the taxonomy of AR patterns, refining existing modules, and conducting field tests to validate the system's capabilities in complex, real-world environments.

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REFERENCES

- [1] P. Fraga-Lamas, T. M. Fernández-Caramés, M. Suárez-Albela, L. Castedo, and M. González-López. A review on internet of things for defense and public safety. *Sensors*, 16(10):1644, 2016. 1
- [2] J. G. Grandi, Z. Cao, M. Ogren, and R. Kopper. Design and simulation of next-generation augmented reality user interfaces in virtual reality. In *2021 IEEE Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*, pp. 23–29. IEEE, 2021. 1